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MASAHIRO TORIYAMA, MOHANBIR SAWHNEY, AND KATHARINE KRUSE

Sony Computer Science Laboratories:

Sustaining a Culture and Organization for Fundamental Research

In late 2019, Dr. Hiroaki Kitano, the president, CEO, and director of research at Sony Computer Science Laboratories (Sony CSL), was considering the future of the organization, a blue-sky* research facility in Tokyo funded by Sony Corporation. It was a job like no other—just as Sony CSL was a place like no other. Kitano, a polymath with expertise in particle physics, robotics, systems biology, and artificial intelligence, had been handpicked by Dr. Mario Tokoro, the founder of Sony CSL. His charter was to lead the organization and its 30 world-class researchers in exploring "wild," cutting-edge scientific research with the lofty mission of improving the future of humanity.

After ten years at the helm of Sony CSL, Kitano had decided he would be stepping down sometime over the next few years. To ensure that Sony CSL would continue to execute on its mission of fundamental research, Kitano wanted to make a transparent plan for the future of the organization before the leadership transition. The plan for Sony CSL's future involved three key decisions. First, what should be the optimal structure and governance for the organization: should

* "Blue-sky" research is driven primarily by the desire to increase scientific understanding, without necessarily having a real-world application.

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it preserve its high level of autonomy, or should it align more closely with Sony's business priorities? Should the scope of its projects, which currently ranged from abstract, fundamental research to start-up incubation, be narrowed? Second, how could Sony CSL scale its impact on Sony and society at large, given its small size? Finally, Kitano had always resisted measuring the performance of Sony CSL with standard key performance indicators (KPIs), reasoning that a metrics-based approach could stifle creativity and blue-sky innovation. However, without measurable KPIs, how could Kitano—or Sony—gauge the strength of Sony CSL's project portfolio or its impact on the future of humanity?

Kitano saw Sony CSL as a unique organization, where "'crazy' people bring their wildest ideas to life."[1] Kitano was very concerned about preserving this culture of innovation, which was central to the mission of Sony CSL. However, such a culture and organization relied heavily on Kitano's own personality and credibility, so it might not be able to withstand Kitano's departure. Making the right decisions on governance, structure, scaling, and metrics would be critical for Sony CSL to continue its mission of pursuing fundamental research.

Overview of Sony

Sony Corporation was founded by Masaru Ibuka and Akio Morita in 1946 as "an ideal factory that stresses a spirit of freedom and open-mindedness, and where engineers with sincere motivation can exercise their technological skills to the highest level."[2]

Sony quickly became famous for its innovative consumer electronics, such as the first transistor radio (1955), the portable television (1960), the Betamax VCR (1975), the Walkman (1979), and the entertainment robot AIBO (1999). (See **Exhibit 1** for a history of Sony's products.) At the same time, Sony diversified into entertainment (music, movies, video games) and financial services (banking and insurance).

In the 1990s and 2000s, competition in consumer electronics intensified, and the rise of Korean and Chinese companies affected Sony's business results. Between 2012 and 2018, CEO Kazuo Hirai led Sony through a turnaround of the electronics businesses, which involved downsizing or selling off poorly performing divisions, such as personal computers, and investing heavily in growing divisions, such as game and network services (e.g., PlayStation) and CMOS image sensors. Sony's 2019 revenue was $77.991 billion, a 1.42% increase over the previous year,[3] and its most profitable divisions in 2018 were game and network services, music, financial services, and semiconductors (see **Exhibit 2**).

Sony CSL

Beginning in the 1980s, Sony's senior leaders began to recognize that the competitive advantage Sony had established in consumer electronics would not sustain the company during the coming decades. Its core strength lay in analog electronics technology, which was slowly being phased out in favor of digital technology, and Sony's engineers were not prepared for the change. The general mood of Sony's engineers at the time was to "stay away from those computer things,"[4]

but that attitude was not shared by electrical engineer Toshitada Doi, who wanted Sony to be at the forefront of technology, like IBM. It would be through Doi that Sony CSL would get its start.

The Founding of Sony CSL

In the mid-1980s, Doi called Mario Tokoro, an associate professor at Keio University who had worked with Doi on Sony projects in the past, with a proposal: "Sony has no computer experts. Would you consider becoming head of a new computer science business unit here at Sony?"[5]

Tokoro offered a counterproposal: he would not come to Sony to lead a new business unit (BU). Instead, he would create a new computer-science research lab that would offer its researchers complete freedom to explore groundbreaking scientific breakthroughs without feeling the pressure to generate profits for Sony. It would be the "ideal lab for creative researchers, much better than university labs."[6] Tokoro emphasized that by heading up this new unit, he had no intention of becoming a corporate drone: "This research institute is . . . not for money-making. It is a research institute for Japan, for the world."[7] Thus, in February 1988, Sony Computer Science Laboratories was born.

The Tokoro Era (1988–2008)

Sony CSL was founded with a clear set of core principles: to be fundamentally innovative and long-lasting; to attract world-class researchers and give them total freedom; to prioritize human decision above rules; and to establish a precedent of cooperation and feedback from researchers across disciplines. (See **Exhibit 3** for a detailed mission statement for Sony CSL.)

Tokoro was determined that Sony CSL should remain small. He proposed that the new institute's budget remain fixed at a constant small percentage (1/10,000) of Sony's revenue. This fixed budget approach allowed Sony CSL researchers to pursue long-range projects without worrying about the short-term impact of their projects on Sony's bottom line. Contributing to Sony's bottom line should be encouraged, Tokoro believed, not expected; if it became Sony CSL's primary obligation, the organization would no longer be fulfilling its mission as a world-class research institute. "An orientation toward sales and profit makes research a slave to money," he said.[8]

The lab was established as a 100% subsidiary of Sony, with Tokoro reporting directly to Sony's CEO. This reporting structure made it unique among research divisions within Sony. The company had more than 20 R&D divisions, but each was domain-specific and reported to a particular BU; for example, the semiconductor BU had its own very large R&D division devoted specifically to innovating and developing technology that improved semiconductor performance and manufacture. Unlike Sony CSL, these R&D divisions had clear performance-related goals directly related to their impact on Sony's bottom line.

To become a world-class research institute, Sony CSL needed to attract and retain the highest quality of researchers. Sony CSL's general policy was to put the researcher first, rather than the

research area. Tokoro did not seek to fill an open position in a specific field of research; the decision to hire was made entirely on an individual basis.

Tokoro had a largely hands-off management style: "KPIs and micromanagement kill creativity," he said.[9] He gave researchers a free hand regarding their theme and process but intensely evaluated their results, wanting them to strive for nothing less than breakthroughs. During Tokoro's tenure as Sony CSL president, projects focused on advances in computing. A sampling of projects included:

- Early augmented-reality technology such as the NaviCam (1994), a handheld computer equipped with a camera that detected real-world environment, enabling the user to view the surroundings with context-specific information provided by the computer, and CyberCode (1996), a visual tagging system in which binary-coded two-dimensional images loaded onto a camera were translated into 3D images that appear to exist in the real world
- The first example of predictive text, POBox (1998), which allowed users to enter text without providing full, complete, correctly spelled information; this technology was later adapted into Sony smartphones
- Distributed object-oriented operating systems such as Aperios (1998), which powered Sony's entertainment robot AIBO

After 20 years of leading Sony CSL, Tokoro decided to step down in 2008. Officially, the head of Sony CSL was supposed to be decided by Sony's board of directors, but, knowing Sony's CEO and board would not disagree with his recommendation, Tokoro simply named a successor: Kitano, a researcher who had been with Sony CSL since 1992. When asked why he chose Kitano to take the reins of the organization, Tokoro recalled that, in addition to Kitano's scientific brilliance, "I thought Kitano had a business mind, so I appointed him."[10]

The Kitano Era (2008–Present)

Though Kitano's doctoral work had been in computer science, he had an extraordinary knowledge of a breadth of disciplines, including physics, biology, and art. (See **Exhibit 4** for details on Kitano's background.) Under Kitano's leadership, Sony CSL began branching out into fields beyond computer science; if computer/AI technology could be applied to a field to make improvements, then Kitano felt it was within Sony CSL's purview and was worth pursuing. In addition, Kitano strengthened the business orientation of Sony CSL. He believed that offering a balanced result—with some projects contributing corporate competence and providing technological support to enhance business opportunities for Sony—would make up for Sony CSL's pursuit of wilder, more long-term, and indirect-benefit projects.

Structure and Governance

Sony CSL remained small and, organizationally, quite simple. (See **Exhibit 5** for Sony CSL's organization chart.) Kitano, as president/CEO of the organization, was the head, and

all 30 researchers reported to him.* In addition to these researchers, Sony CSL had a group of nonscientists whose role was to connect the research Sony CSL was doing to real-life business applications, investigating opportunities for profitability both internally at Sony and externally via technology transfer and licensing. Sony CSL also employed a small administrative-support team.

Kitano believed that decisive decision-making and central direction from a leader with a deep understanding of the research was crucial for Sony CSL's success.[11] His formal review process for researchers was simple. Every March, each researcher had to submit to Kitano a report that would detail the past year's progress on his or her project. In June, Kitano would meet with each of the researchers for one-on-one interviews, during which they would discuss the year's results and together formulate a plan for the coming year. The goals of each project, with criteria for success, were agreed upon in this annual meeting. If Kitano felt that a project was not making adequate progress or had dim prospects, he could terminate the project—which in some cases meant an end to the researcher's employment with Sony CSL.

In addition to receiving one-on-one feedback and direction from Kitano, the researchers made presentations to their colleagues twice a year, in April and October. All Sony CSL researchers were expected to attend this two-day offsite meeting and to critique their colleagues' work. Kitano was front and center at these meetings, challenging the research presentations to hammer out the weak spots in the work and to ensure all alternatives had been considered.

Like Tokoro, Kitano was disdainful of key performance indicators and standard metrics when assessing the performance of his researchers:

> *Research, for a researcher, is his entire life, and [Sony CSL's research] is so diverse that it is impossible to evaluate with a uniform KPI. KPIs are not a good motivator for a true researcher. Given the high caliber of Sony CSL's researchers, if I defined some KPIs they would achieve them for sure. Then achieving KPIs would be the goal, rather than pursuing the true purpose and value of their research. But some sense of tension is necessary; we have a research presentation meeting which all researchers must attend. Even though the disciplines are different, the quality of work can be judged by each other quite precisely. No one wants to be judged as inferior in quality. The questions are surprisingly sharp and relentless. There is no researcher who cannot evaluate quality, nor any researcher who cannot bear those sharp questions, in Sony CSL. What I believe to be most important are the vision, imagination, and passion that drive researchers and their capability to strategically frame the project and execute it fast and solid. Most conventional KPIs, such as publications and patents, are mere byproducts or tools to accomplish the vision.*[12]

Kitano's own reporting cycle was even simpler. Every February, Kitano met with Sony's CEO and CFO to deliver his annual review; during this meeting, Kitano reported his fiscal plan

* Eight of Sony CSL's researchers worked in Paris and reported to their own director, who in turn reported to Kitano. Although Kitano did not provide the same level of close attention and individual feedback/guidance for the Paris researchers, he received reports from the Paris director and visited frequently to check on the projects' progress and speak with the Paris researchers in person.

for the next year, and the budget was formally approved. This was the only formal review of Sony CSL's activities.

Culture

In a typical university lab environment, researchers tended to communicate, collaborate, and seek feedback only with and from others in the same field, usually speaking with people who were either their subordinates or superiors in a rigid hierarchy. The culture at universities was therefore often closed and siloed.

By contrast, Sony CSL was a relatively flat organization, with each researcher representing a unique discipline or set of disciplines. When researchers sought feedback from their peers at Sony CSL, therefore, they were obtaining opinions from experts in a variety of fields, enabling cross-pollination across domains and providing crucial insights that a "naïve expert" can bring. Sony CSL had an extremely open culture, reinforced by Kitano and exemplified by the rigorous semiannual all-hands presentations; the researchers were expected to mutually exchange feedback, suggestions, or even harsh criticism in frequent, honest interdisciplinary discussions.

"Act beyond borders" was the guiding principle Kitano established for the lab when he assumed leadership. He believed the problems Sony CSL was tackling were tangled up in complex, open systems and consequently demanded solutions that crossed traditional borders of academic disciplines.[13] Further, it would not be sufficient to merely write a research paper and expect that someone, somewhere would eventually turn the research into an actionable real-world solution. Action was needed to achieve Sony CSL's mission of contributing to the betterment of humanity; thus, the "act" was as critical as the "beyond borders."

One researcher described the culture as complete freedom—but with a catch: "I like the freedom, and I dislike the freedom. It is scary because no one tells you anything. You do not even have to be there. There is nothing to stop you from being lazy. [But] before the biannual reporting event, the pressure becomes maximum. As for the quality, everyone is relentless."[14] Not all researchers flourished in Sony CSL's environment of continual open critique; one ex-researcher described Kitano's management style as dictatorial.[15]

Routes to Impact

Unlike the R&D labs at Sony, which were created to develop solutions to well-defined problems, Sony CSL tended to explore the unknown, often with no proven business application but with the belief that if the research were valuable, it could transform the world into a better place and would eventually trickle down to have an impact on Sony's bottom line. Sony CSL strictly defined its research as a "one way" route to development; that is, the ideas began with the researchers—never Sony.[16] As Kitano explained:

> *One of the benefits of having an organization like Sony CSL within Sony Group is it embraces a scope and time scale far beyond Sony's current business. Sony CSL's work might be able to provide Sony readiness for unexpected situations or future business development. If we are*

> *working on the predictable areas that are relevant to Sony's current business, then we are not doing our job. If we started listening to development engineers' needs, which fall within that predictable scope, our research would become subcontracting of BUs, which look three to five years into the future at most; we must avoid it by all means.*[17]

Further, Sony CSL was strictly about research—never development. Its researchers were free to explore ideas without concerning themselves with whether the technology had a specific commercial use.

Despite not having an explicit directive from the C-suite to improve Sony's bottom line, Tokoro recognized the potential impact that Sony CSL's work could have on Sony's business. He had thus created a Technology Promotion Office (TPO) within Sony CSL to assist in technology transfer to Sony. When Kitano assumed leadership, however, he felt pressured by the rapid advance of technological development and increased competition to maximize Sony CSL's impact in as many ways as he could, including to Sony's bottom line, if possible.

When researchers joined Sony CSL, they could pursue their choice of project; the main criterion was that the research should use technology in a way that would ultimately benefit humanity on a grand scale. (Notably, however, Kitano also supported research pursued purely out of curiosity, driven by a researcher's passion and imagination.) With Kitano's approval, Sony CSL's researchers embarked upon a wide variety of projects, such as the invention and study of new academic disciplines, AI-powered technological advances that augmented the performance and perception of the human body, and new agricultural methods that spurred biodiversity and reversed ecological damage. (See **Exhibit 6** for an overview of Sony CSL projects, and see **Exhibit 7** for details on a few projects.)

To balance freedom with business impact, Kitano felt Sony CSL needed to establish several routes to impact that would allow the researchers the freedom to explore while also not completely ignoring the business needs of the corporation that funded them nor of third parties that might find the research valuable. The TPO that had begun during Tokoro's era of leadership was the route to impact that had the most direct connection to Sony's BUs. The office's primary priority was to find an opportunity to transfer technology from Sony CSL to a Sony BU so it could be used to develop or improve a product or process. Many TPO staff had come from elsewhere within the company and thus had strong networks within Sony. They were not scientific experts but learned enough through frequent, in-depth interviews with the researchers to find connections between the research and Sony's business and, upon discovering a possible match, to pitch the technology to BU leaders within Sony. For example, the TPO was instrumental in leveraging Hideki Takayasu's work in econophysics to improve the yield rate for Sony's semiconductors (see Exhibit 7 for more details).

Sometimes Sony CSL's research might be a good fit for Sony's businesses but had no immediate application because it was too forward-thinking. It might be too early for the market, or perhaps the necessary complementary technology had not yet been invented. In these cases, the TPO put the technology on ice, as it were, saving the source code, hardware, demo video, and/or documentation in a way that would enable future team members to "thaw out" the research when

it would be more useful. The TPO never took requests from the BUs for new projects, so Sony CSL could maintain its appropriate focus on very-long-term, unpredictable areas of research and avoid becoming a subcontractor for the BUs.

Kitano was also asked by Sony CEO Kazuo Hirai to incubate spinoffs in new business domains, a function that eventually was merged with the TPO. The incubation program resulted in spinoffs such as Sony Global Education (a fully owned arm of Sony Corporation) and Ghelia Inc. (an AI startup for industry and business sectors in which Sony owned a minority stake).

Another way Kitano found a real-world impact for Sony CSL's expertise within the larger organization was through the creation of joint labs. These programs contributed to Sony's corporate competence and supported the strategic initiatives of Sony's BUs. One such example was the Data Science Laboratory, which Kitano created so that Sony CSL researchers could provide educational programs in data science to employees across Sony Group, and could support selected projects that benefited from their knowledge.

If Sony CSL's research had no direct application for Sony or the TPO was unable to spark interest within Sony's BUs, then it was looked at by the Technology Promotion and Deployment Group (TPDG). This group's primary responsibility was to pursue a route to impact by building relationships with external partners, either by:

- *Licensing:* monetizing Sony CSL research by licensing the technology to an outside organization
- *Partnerships:* collaborating with external companies (ideally, those in different areas of business than Sony) or organizations (government sectors, NGOs, or international organizations) to develop a new product; the researchers were involved with the process, and the deal was not pursued if they felt the proposed product would be a bad fit for their technology

One such partnership that the TPDG supported was the Small Optical Link project, in which Sony CSL researchers worked with the Japan Aerospace Exploration Agency (JAXA) to develop a long-distance laser communication system for use on the International Space Station (see Exhibit 7 for more details).

If Sony CSL's research turned out to be totally unrelated to Sony's business, the route to impact would be to seek potential benefits for the world or human race in general.[18] In some cases, researchers themselves might become social entrepreneurs or work with such a person to fulfill their vision. For academic and social impact projects such as these, Kitano formed the Research Activation Group, which aimed to disseminate basic research conducted at Sony CSL and to form external collaborations that would change people's perception, transform society, or create new academic or artistic fields. One such example was Masatoshi Funabashi's Synecoculture™ project, which sought to optimize agricultural production through insights derived from complex systems science (see Exhibit 7 for more details).

The various routes to impact allowed Sony CSL to pursue a wide range of outcomes for its research, ranging from direct benefit to Sony's BUs to licensing, partnerships, and contributions to academic research and society.

Fundamental Research at Other Organizations

Sony was not the only company that funded exploratory, fundamental research that might have no immediate business applications. Kitano could look to other corporate research organizations as he tried to define the ideal structure and governance for Sony CSL. (See **Exhibit 8** for a comparison between Sony CSL and a few of these other research labs.)

IBM Watson Research Center

Founded in 1945, the Watson Research Center was the first pure science research lab operating within a corporation in the United States. The center was the headquarters of IBM Research, an enormous organization comprising more than 3,000 employees and spanning 12 different laboratories on six continents.[19] During its early years, the center consisted of a small group of young scientists who used IBM's advanced computers to research topics such as atomic fission and the moon's orbit. But by the 1990s, the company's research operations had become bloated, slow, and expensive.[20] IBM Research was granted more patents per year than any other company in the world, but managers had a difficult time launching and growing new businesses that would commercialize these inventions and exploit growth opportunities arising in the marketplace.[21] IBM employees were risk-averse, reluctant to give up historical products and services that were proven successes. Innovations that did not naturally fall within an existing line of business did not get developed. To address this concern, IBM began designating its projects either Horizon 1 (those with an application in a mature, established line of IBM's business), 2 (those with an application in one of IBM's growing businesses), or 3 (areas known as "emerging business opportunities," those with less well-defined or speculative applications). H3 projects, as the future of the company, began to be managed differently than the other two, with a different leadership style and a performance evaluation based on project milestones rather than pure P&L. A three-color system was also established to measure progress: red indicated serious problems with the project; yellow meant limited progress or issues that needed be resolved; and green signified sustained success.[22] Though these projects were given more flexibility than H1 or H2 projects, IBM still imposed limits: if an H3 project did not develop a sustainable revenue stream within ten years, IBM most likely would shutter it.[23]

X

A more contemporary example of fundamental research was X (formerly called Google X), founded in January 2010 as a semi-secret R&D division within Google. After a 2015 corporate restructuring, Google X was spun off into a separate subsidiary under the umbrella of Google's new parent company, Alphabet, Inc., and was renamed X. The head of X, Astro Teller, saw the lab as living in the space between pure research (high-risk, ideals-oriented) and pure corporate (low-

risk, profit-oriented), but he noted that "being a 'corporate lab' is a difficult balancing act: place big bets on the future, but don't spook the people giving you the money."[24] X brought together a diverse group of inventors and entrepreneurs to create a 10x impact on the world's most intractable problems. X focused on projects that demanded the aspiration and riskiness of fundamental research with the speed and ambition of a startup. Its projects included Waymo (self-driving cars), Tidal (underwater sensing and perception), Verily (sensors and software for life sciences), and Makani (energy kites to generate energy from wind). X encouraged its researchers to identify risks early by testing the most challenging aspects of their concepts as soon as possible in the R&D process. Several X projects became separate business entities under the Alphabet umbrella, but overall, X was not a revenue generator for the organization; no blockbuster commercial product had yet emerged from X's factory.[25] At the end of 2019, X was brought back under the Google umbrella when Sergey Brin and Larry Page stepped back from the day-to-day management of the company they had founded, naming Sundar Pichai as CEO of both Alphabet and Google. This move to fold X into Google was a tacit admission that the attempt to create business discipline for fundamental research had failed.

MIT Media Lab

Founded in 1985, MIT's Media Lab was an unusual hybrid of academic program and research-focused lab. It took an actively "antidisciplinary" approach to its research, seeking out ideas that fell into "white spaces" between fields of study—the areas that would traditionally be most difficult to acquire either federal or private funding to explore.[26] The lab was 80% funded by corporations, which in exchange received membership in a consortium, giving them access to interactive demonstrations, opportunities to recruit the lab's students, opportunities for cross-company collaboration and networking, and, perhaps most important, opportunities for commercialization of the Media Lab's research. Any IP resulting from Media Lab research was shared among all corporate members, under a non-exclusive, royalty-free license agreement, in perpetuity.[27] The funding itself was not designated for a specific purpose, which allowed the corporate members to absorb the risk for the most experimental of Media Lab research.

In practice, however, the Media Lab's patents were rarely commercialized. A former director saw the Media Lab's role as one of pure discovery, not problem-solving. "Once we have done that discovery and development, our member companies take those inventions and commercialize them. When other people start getting into an area, we try to get out."[28]

The Path Forward

Kitano wanted to specify a plan for Sony CSL that would secure its future within Sony and as an innovative force in the world at large. He wanted Sony CSL to outlive him and sustain its unique culture. Kitano identified three dimensions along which he needed to outline the future of Sony CSL: structure/governance, scaling social and business impact, and measuring performance.

Structure and Governance

Sony CSL operated autonomously within Sony, with Kitano overseeing every aspect of its operation. Sony CSL's small size and budget meant it functioned primarily on trust between Kitano and Sony's CEO. Kitano had total authority over the project portfolio and its management; he decided whether his researchers were on-target with their projects and whether they were adding enough value to Sony CSL and its parent company. Kitano thought of himself as a trusted advisor who had the CEO's ear on long-term strategy:

> *Although Sony has 100% share of Sony CSL and we report to the CEO, we never thought that we were in a master-servant relationship; rather, our relationship is an equal partnership. Often, we are asked if Sony CSL is Sony's Corporate Social Responsibility activity, which is not correct. While we have projects aiming at high social impact, Sony CSL does contribute to Sony's business in a big way. For example, Sony's most profitable business is the semiconductor, for which a higher yield rate is critical. Our researcher Dr. Takayasu and his team, in partnership with the semiconductor division, are constantly improving the yield using an advanced statistical and machine learning approach. Two years ago, we created the Data Science Laboratory (DSL) to provide educational programs in data science across Sony Group as well as to support selected projects of high strategic importance. We have been doing unique research on music, which is now creating new frontiers for the music business. Our global sustainability projects in agriculture and energy are gaining importance, as Sony Group is paying more attention to Sustainable Development Goals (SDGs). Recently, Sony decided to create a new organization, Sony AI, focused on AI technologies strategically aligned with Sony's current and future core businesses. I am leading the effort with one of Sony CSL's researchers, Dr. Michael Spranger. Starting an organization like Sony AI within a relatively short period of time would not be possible without the expertise and talent pool at Sony CSL. I think we are providing Sony with readiness for the future.*[29]

The governance of the organization, meanwhile, was implicit. Kitano reported directly to the CEO. He also reported results to Sony CSL's board of directors (but not to the board of directors of Sony Corporation). However, Kitano had extensive informal discussions and dialogues with a broad range of C-suite officers within Sony Group. Whereas the CEO needed to answer to Sony's shareholders, Kitano primarily felt a responsibility to Sony CSL's mission and researchers; if the CEO was satisfied with his annual review, Kitano would continue acting according to his own best judgment. Knowing he had only the CEO to satisfy, Kitano always made sure, in his annual review, to mix Sony CSL's grand vision with its practical contributions to Sony Group. He first mentioned Sony CSL projects that had considerable impact on Sony's business, such as improved yields for semiconductors, then transitioned to explaining the "wilder" projects. Explained Kitano: "This way, I can make sure Sony CSL is not viewed as a charity but an organization with strategic importance for Sony's business. Then, we can move to explain blue-sky projects in a relaxed mood rather than one of scrutiny, where the CEO and CFO are enjoying the intellectual conversation, and we can have a constructive and informative discussion."[30]

An autonomous structure and implicit governance had worked for Sony CSL thus far, but it was risky because it relied entirely on a handshake between two executives. If a new CEO came

on board, he or she might not value exploratory research to the same extent, or place restrictions on the scope of Sony CSL's research. If Sony's business dipped, Sony CSL could be faced with restructuring or budget cuts, especially if Sony CSL had grown. At the other end, Kitano's successor might not be able to balance benefits to Sony with benefits to external parties. As Sony CSL expanded its operations to range from fundamental research (e.g., expanding scientific understanding in agriculture, biology, energy, mathematics, AI, human-computer interactions, music, and interactive art) to business incubation (e.g., space-based communications systems or financial services), finding a single individual who would be capable of leading the whole spectrum of the lab's activities would be a major challenge. Kitano thought the organization needed to be reshaped to be more robust and easier to manage rather than depending on the capability of a single individual to cover this breadth of activities. Obviously, this transformation would require redefining organizational structure and governance.

Perhaps, Kitano considered, the governance of Sony CSL should become more explicit, for instance, by establishing a supervising/advising board, along the lines of the membership consortium of MIT's Media Lab. An advisory board could either preserve Sony CSL's current autonomy or increase its connection to its parent organization. On the autonomous end of the governance spectrum, the board could be composed mainly of external parties, such as ex-presidents of Sony CSL (i.e., Tokoro and Kitano) and other experts and visionaries, with only the CEO representing Sony's interests. A board that preserved Sony CSL's autonomy carried the risk, however, that Sony CSL's activities would veer too far away from Sony's core activities. On the connected end of the governance spectrum, the board could consist of a mix of BU directors and Sony Group heads, to forge a closer relationship between Sony CSL and Sony BUs. But Kitano worried about Sony CSL's research becoming biased by the BU priorities and losing its creativity—the main reason for its existence.

Another issue to consider, if a board were instituted, was the level of control such an entity might have over Sony CSL operations. On the weaker end of the level of control, the board would function only in an advisory capacity to provide councsel and feedback for Sony CSL's president. On the stronger end of control, the board would have the power to nominate and vote on Sony CSL's president, approve or reject strategic plans, or even influence the direction and themes of research.

Scaling Business and Societal Impact

Sony CSL was not operating at a loss. Some of the projects that had begun in Sony CSL's labs had moved to Sony BUs and were now generating money for the company—for example, the millions of dollars that econophysics research had saved Sony through improved semiconductor performance. Kitano knew the financial benefit that Sony CSL had provided Sony could have covered the entire cost of Sony CSL's operations. He felt this gave him some leverage to argue in favor of expanding the group. At the same time, Kitano understood such contributions were not explicitly listed on Sony's balance sheet, as Sony CSL was not directly selling products and services. Thus, unrestricted expansion would be risky if the business climate changed or a new CEO thought a bigger budget for Sony CSL was problematic.

Sony CSL's small size had allowed it to operate without scrutiny from Sony's C-suite and the kind of micromanagement and focus on bottom-line results that would counteract the researchers' creativity. Its small size had also enabled Kitano to provide detailed, one-on-one feedback and direction on every project in the portfolio. But he knew the global impact of Sony CSL was significantly limited, with only 30 researchers at its disposal. Adapting a geopolitical concept, as seen in the US Department of Defense's concept of global power projection, Kitano defined the ideal state for Sony CSL's level of global influence as "being able to apply all or some of its elements of organizational power to rapidly and effectively deploy and sustain influence in and from multiple dispersed locations and domains to respond to crisis, to contribute to sustainability, and to enhance civilization."[31]

Although this definition of impact—maximizing Sony CSL's global influence—was powerful and elegant, it was not easy to operationalize. Kitano knew it would not be as simple as adding researchers. Scaling the impact of Sony CSL was not the same thing as scaling its size.

One approach to scaling impact, he thought, would be a "federated geographical model," which would involve an expansion of Sony CSL by creating several labs in different locations around the world, each patterned after the original Sony CSL. These geographically distributed labs would allow Sony CSL to take advantage of local talent around the globe and to gain local market and customer insights while preserving its proven formula of minimal rules, power concentrated in a president who would be able to have deep discussions on research with each of his reports, and hiring only the "right kind" of researchers (i.e., those with intrinsic motivation and a grand vision). In this model, Sony CSL Tokyo would function as the equivalent to IBM's Watson Research Center, the corporate headquarters of a federation of small research labs.

The federated geographical model had several appealing characteristics. Sony CSL could create centers of research excellence that leveraged local research specialties. For example, it could build a lab in the Boston area that focused on biotechnology and life sciences; one in Tel Aviv with a focus on security; and another in Shenzhen with a focus on manufacturing. Having local facilities would allow Sony CSL to attract world-class talent without those people having to relocate to Japan. In addition, operating in different locations would provide Sony CSL with access to local market needs and insights; currently, its researchers had excellent insight into Japan's problems, but Sony CSL aimed to solve global problems.

However, this model also presented significant drawbacks. Researchers would no longer be co-located, so it would be more difficult to cross-fertilize ideas and provide feedback on research occurring outside one's chosen field. The wilder, antidisciplinary approach of MIT's Media Lab was exactly why Tokoro had founded Sony CSL in the first place: to escape the rigid silos of academia. Kitano wanted to avoid establishing labs that would function independently and in isolation, as he feared this would reduce the opportunities for naïve experts to provide breakthroughs or insights from other fields.

Another approach to scale impact could follow the example of the Media Lab more closely by creating a few clusters of deep research expertise. Sony CSL could organize researchers into "research programs" to allow individual research projects to enter a phase of scalable deployment.

Researchers in a research program would collaborate with multiple stakeholders, such as engineers, software developers, or research fellows. Each research program would be defined loosely by theme, not discipline—for example, communications-related projects. This would allow each program to function independently while also encouraging cross-fertilization across disciplines. Fundamental research was antidisciplinary; it operated in the realm of the serendipitous, and cross-fertilization was an essential ingredient. As Sony CSL's own history had shown, the mingling of two seemingly unrelated disciplines, economics and physics, had produced its most profitable application for Sony. Research programs would enable Sony CSL to increase the depth of its expertise in specific domains and make deployment more scalable.

As the research programs coalesced, team leaders could be appointed to manage them. Recruitment, tracking, and individual evaluations at Sony CSL would cease to be dependent on a single individual (i.e., Kitano) and would be dispersed among the leaders of the research groups, along the lines of a department chair at a university. Having team leaders would also serve as an incubator for future management roles, thereby ensuring Sony CSL had a viable succession plan once Kitano was gone. However, adding a layer of management would mean governance would have to become more explicit and codified to keep all groups unified and managed consistently. This option would also likely need to involve hiring more administrative staff to identify and coordinate external stakeholders.

Measurement and Rewards

The criteria by which Sony's R&D labs were judged were concrete—papers, pilots, and projects—and thus determining success and compensation were straightforward. In the exploratory research environment of Sony CSL, however, measuring success could be tricky. Because the mission was so grand—effecting large-scale change on a global scale—the extent to which a researcher succeeded might not be clear until decades later. During his tenure as Sony CSL's president, Kitano had shunned concrete milestones and KPIs; Kitano himself was the ultimate arbiter of a project's success or failure, and Sony's CEO trusted Kitano's judgment. The criteria were contextual, variable, and highly dependent on personal relationships. Given the close tie between metrics and compensation, however, the lack of concrete metrics might translate to a lack of incentives for researchers—or even the future Sony CSL leader—to perform.

Kitano wondered whether his successor should continue to be solely responsible for defining success and failure of a project, or whether success should begin to be defined by the researchers themselves, or even defined at the corporate level—if not entirely, then at least to some extent. As with other factors that would endeavor to maintain the organization's status quo, leaving all the responsibility with Kitano's successor was risky. This was because it depended entirely on who would be sitting in the president's chair; how could any successor reasonably be expected to be an expert in every field that Sony CSL researchers studied?

Kitano wondered if he could create a standard set of criteria that could be used to evaluate all the projects in Sony CSL's portfolio. Following IBM's example, Sony CSL could evaluate progress on a system of milestones rather than P&L, perhaps with a similar simple three-color scoring

system that indicated the project's health. If Sony began establishing some standardized milestones for the lab, it would allow transparency for the success or failure of a project, and as with X, might encourage "fast failure"—essentially giving researchers the freedom to shutter a project and move onto something more promising rather than waste ten years on something that would never be successful. Defining a standard set of criteria that could be applied across such a wide variety of projects would be a challenge, however. How long should an idea be given before it must bear fruit or be terminated? Further, Kitano was worried about stifling Sony CSL's atmosphere of freedom to create and ideate. Establishing a standard set of criteria to measure all projects equally might be impossible, given the wild variety of Sony CSL's portfolio.

Perhaps, he thought, it could be done on a project-by-project basis, with the researchers themselves defining success. Each researcher could sketch out the stages of their project, from ideation to maximum impact, defining the success criteria at each stage. When the project was funded, this description of stages and success criteria would be agreed upon by both the researcher and the president. The terminology used to describe the stages and criteria could be somewhat standardized—for example, impact on Sony's revenue, media mentions, bibliographic citations—so that the strength of the overall portfolio became evident, as every project could be compared to the others. These metrics could be designed to evolve. This approach had the benefit of asking researchers to think out the full impact of their research. However, the cost of this approach would be the significant burden on each researcher to plan these criteria; it might even necessitate hiring extra staff to produce this documentation.

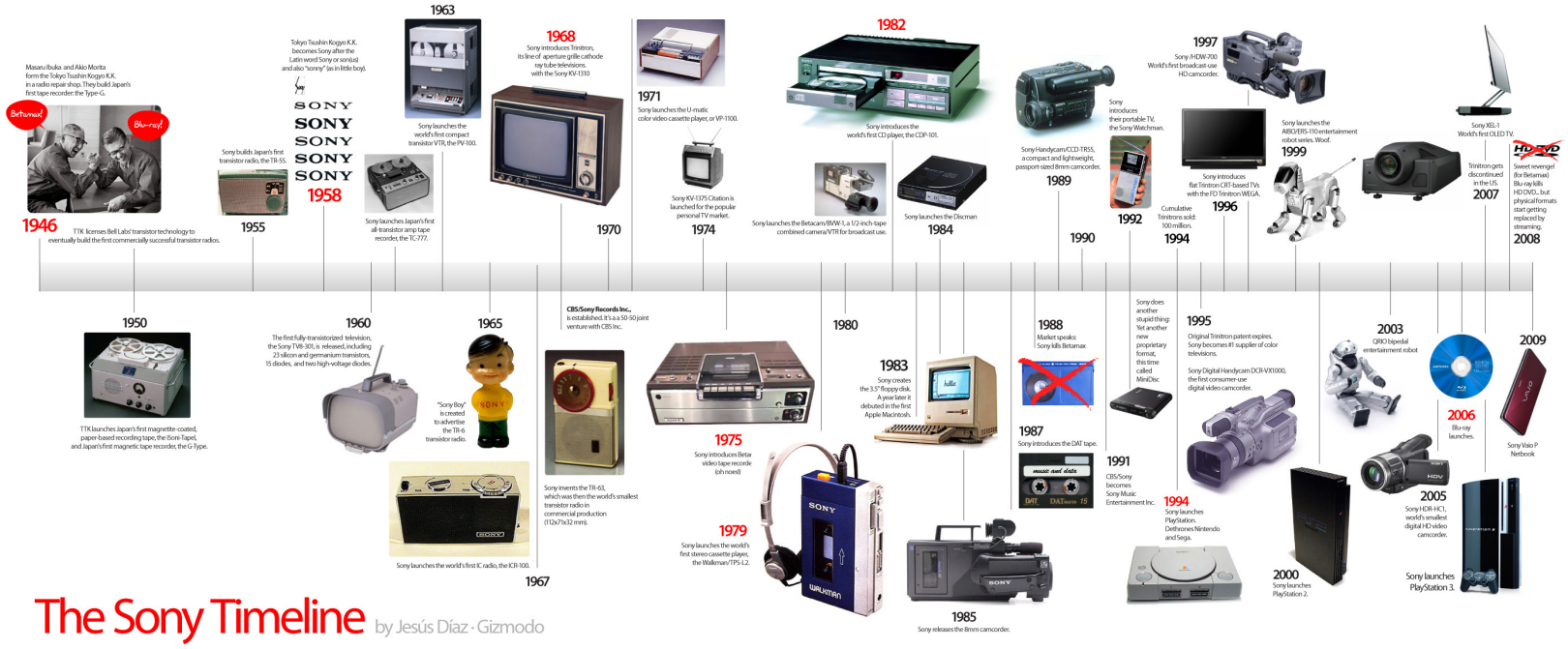
The uncertainty surrounding metrics extended to the performance of Sony CSL's portfolio. Most of Sony CSL's research would not likely fall into the "superstar" breakthrough category. Assuming that a successful outcome meant either increased revenue/savings for Sony or a major impact on the world, the next president of Sony CSL might only need one such success to justify Sony's continued investment in Sony CSL (and to justify the CEO's faith in Kitano's successor as a leader). If, however, the portfolio contained only moderately successful projects—those with only a marginal benefit to Sony BUs, an incremental innovation to an existing product, or technology that fell somewhat short of expectations—then it would become more difficult to justify Sony CSL's existence, and trust in the new president's leadership might be shaken. But if a breakthrough project would not generate business impact for ten or more years, how could the new president's performance be evaluated annually?

One approach might be to hold an evaluation for the president only after a certain number of years and have it done by former Sony CSL presidents, board members, and Sony Group representatives. In the interim, the president would have sole discretion over hiring and allocation of resources. Incentives and compensation would be based on the final outcome, not measured along the way. Evaluating performance in such large time increments would deliver a clear message to Sony CSL's president about the scope of Sony's expectations for his lab's performance and the investments he made in the researchers and their projects. See **Exhibit 9** for hypothetical examples of different approaches to performance review and evaluation of the Sony CSL president and of a Sony CSL researcher.

Conclusion

Kitano was ready to move on from Sony CSL but knew he couldn't leave the organization without a plan for the future—one that would preserve his and Tokoro's legacy and continue the central mission of innovative research that would benefit humanity. The three dimensions of structure and governance, scaling impact, and measuring performance were interdependent; a change in one dimension would bias decisions in the others. Kitano's decisions in these three areas had to be consistent with his vision of the ideal research environment to ensure Sony CSL could maximize its impact on society. Above all, he knew he had to prepare the Sony CSL for the future while preserving the "soul" of the organization that made it a unique and special place.

Exhibit 1: History of Sony Products



Source: Jesús Diaz, "The Sony Timeline: Birth, Rise and Decadence," Gizmodo, March 4, 2010, https://gizmodo.com/the-sony-timeline-birth-rise-and-decadence-5485809.

Exhibit 2: Sony Group Sales and Operating Income by Segment

FY2018 Results by Segment

(Bln Yen)

		FY17	FY18	Change	FX Impact
Game & Network Services (G&NS)	Sales	1,943.8	**2,310.9**	+367.1	-9.4
	Operating income	177.5	**311.1**	+133.6	-4.8
Music	Sales	800.0	**807.5**	+7.5	+0.6
	Operating income	127.8	**232.5**	+104.7	
Pictures	Sales	1,011.1	**986.9**	-24.2	+3.7
	Operating income	41.1	**54.6**	+13.5	
Home Entertainment & Sound (HE&S)	Sales	1,222.7	**1,155.4**	-67.3	-24.6
	Operating income	85.8	**89.7**	+3.8	-21.6
Imaging Products & Solutions (IP&S)	Sales	655.9	**670.5**	+14.6	-3.7
	Operating income	74.9	**84.0**	+9.1	-3.2
Mobile Communications (MC)	Sales	723.7	**498.0**	-225.7	-4.9
	Operating loss	-27.6	**-97.1**	-69.5	+2.0
Semiconductors	Sales	850.0	**879.3**	+29.3	+0.1
	Operating income	164.0	**143.9**	-20.1	-0.5
Financial Services	Revenue	1,228.4	**1,282.5**	+54.2	
	Operating income	178.9	**161.5**	-17.5	
All Other	Sales	407.2	**345.7**	-61.4	
	Operating income	-23.5	**-11.1**	+12.4	
Corporate and elimination	Sales	-298.8	**-271.0**	+27.8	
	Operating income	-64.1	**-74.7**	-10.6	
Consolidated total	Sales	8,544.0	**8,665.7**	+121.7	
	Operating income	734.9	**894.2**	+159.4	

· Sales and Revenue in each business segment represents sales and revenue recorded before intersegment transactions are eliminated. Operating income in each business segment represents operating income reported before intersegment transactions are eliminated and excludes unallocated corporate expenses (applies to all following pages).
· Both Sales and Revenue include operating revenue and intersegment sales (applies to all following pages).
· For further details about the impact of foreign exchange rate fluctuations on sales and operating income (loss), see Note on page 29 (applies to all following pages).

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Source: Sony Investor Relations, Earnings Releases, FY2018 Presentation, https://www.sony.net/SonyInfo/IR/library/presen/er/archive.html.

Exhibit 3: Sony CSL Mission Statement and Goals

Research for the Future of Humanity

Research opens up the future. And so this future first exists in the imaginations of researchers. Sony Computer Science Laboratories' (Sony CSL) support of researchers with extraordinary imaginations is what makes it a place that is helping to create a future beyond most of our wildest dreams.

We declare that we conduct research for the future of humanity. Humankind has astonishing potential and great diversity. Bringing out people's maximum creativity, intellectual power, and physical capabilities, and then using technology to go even further, is how we will continue to unlock humanity's potential. People using technology to extend their abilities could be thought of as the next stage of our evolution, with the potential to make significant and rapid leaps forward.

Humanity has this great potential, but it is also confronting exceedingly difficult problems related to the environment, poverty, aging, food, health, medicine and more. These problems are complicated as well as diverse—we are going up against an extremely harsh reality. In order to solve these problems, I believe we will need entirely new ideas that go beyond borders, and exceptionally effective ways of putting these ideas into practice.

On our 20th anniversary, we at Sony CSL proposed moving from an approach to science based on closed systems to an approach based on open systems. We called it Open Systems Science. Open systems are systems made up of complex, interacting elements, with boundaries that cannot be clearly defined; global society and the environment are two textbook examples. When we at Sony CSL come up with a creative idea, we are not boxed into limited knowledge networks. Moreover, a humanity that has extended its abilities with technology is itself an open system, connected to both natural and social structures.

With its focus on complexity, openness, and uniqueness, open systems research could also be described as an attempt to tackle the challenge of engineering extremely important but hitherto unreproducible phenomena. This sort of research requires a combination of empirical science, *scientia*, and art in its original sense of *ars*, skilled craftsmanship, redefined in the modern context.

In order to solve the problems that naturally arise when attempting to create such a system, we must integrate knowledge from a wide range of domains, and at the same time go beyond academic research and actually head into the field to put our ideas into practice—in some cases, we must continue our research whilst engaging ourselves in this system reform.

This line of thought led us to the Sony CSL motto: "Act beyond borders." We feel it is our duty to solve problems, open new possibilities, and offer something to the world by transcending borders between nations, between scientific fields, and between science and business. And crucial to this is the imagination and drive of our individual researchers. In a sense, you could say that each of our researchers is battling the very limits of his or her imagination. People have asked us if these wild ideas of ours will ever come true. Can we imagine the things that will happen,

and the things that will be needing to be achieved, in as much concrete detail as we can? Do we have the ability to make those things happen? Our picture of the future determines the actions we take today.

Sony CSL supports the process of making these wild imaginings a reality. The research done at Sony CSL has created new scholarly fields, made major scientific contributions, and so on. In that sense, it has affected the world. But there are many things (we are working on) that will require concrete action and development within our societies if they are going to truly impact the world. Some of these will enter the world via Sony Group businesses or with companies beyond Sony Group, others will be created through cooperation between public utilities and government entities both domestic and international, others still will be spearheaded by Sony CSL researchers themselves creating new companies. Our researchers have many different methods at their disposal, and they can choose the one that best suits their desires and the nature of their project.

Sony CSL is a framework to make the wildest ideas come true, for the future of humanity.

Source: Hiroaki Kitano, "Research for the Future of Humanity," Sony Computer Science Laboratories, accessed February 26, 2020, https://www.sonycsl.co.jp/about/message.

Exhibit 4: Hiroaki Kitano

Hiroaki Kitano received his BA in physics from International Christian University, Tokyo, and PhD in computer science from Kyoto University. From 1988 to 1994, he was a visiting researcher at Center for Machine Translation at Carnegie Mellon University. His research career includes Project Director of Kitano Symbiotic Systems Project, ERATO, Japan Science and Technology Corporation (1998-2003) followed by Project Director of Kitano Symbiotic Systems Project, ERATO-SORST, Japan Science and Technology Agency (2003-2008). He was involved in establishing the Okinawa Institute of Science and Technology Graduate School (OIST), where he has served as a professor since 2000. He is a Founding President of the RoboCup Federation. He served as visiting professor at several universities, including as a Special Professor at the University of Amsterdam and Sir Louis Matheson Distinguished Professor at the Australian Regenerative Medicine Institute, Monash University, Australia. He has also served as a member of various councils of the World Economic Forum (complex systems, energy, AI and robotics, and quantum computing). He received the Computers and Thought Award from the International Joint Conferences on Artificial Intelligence in 1993, Prix Ars Electronica 2000, Design Award 2001 from Japan Inter-Design Forum, Good Design Award in 2000, 2001, 2010, and 2019, and Nature Award for Creative Mentoring in Science (2009 Japan Mid-career Award), as well as being an invited artist for Biennale di Venezia 2000 and Museum of Modern Art (MoMA) New York in 2001. Kitano was appointed a Senior Vice President of Sony Corporation as of June 2016.

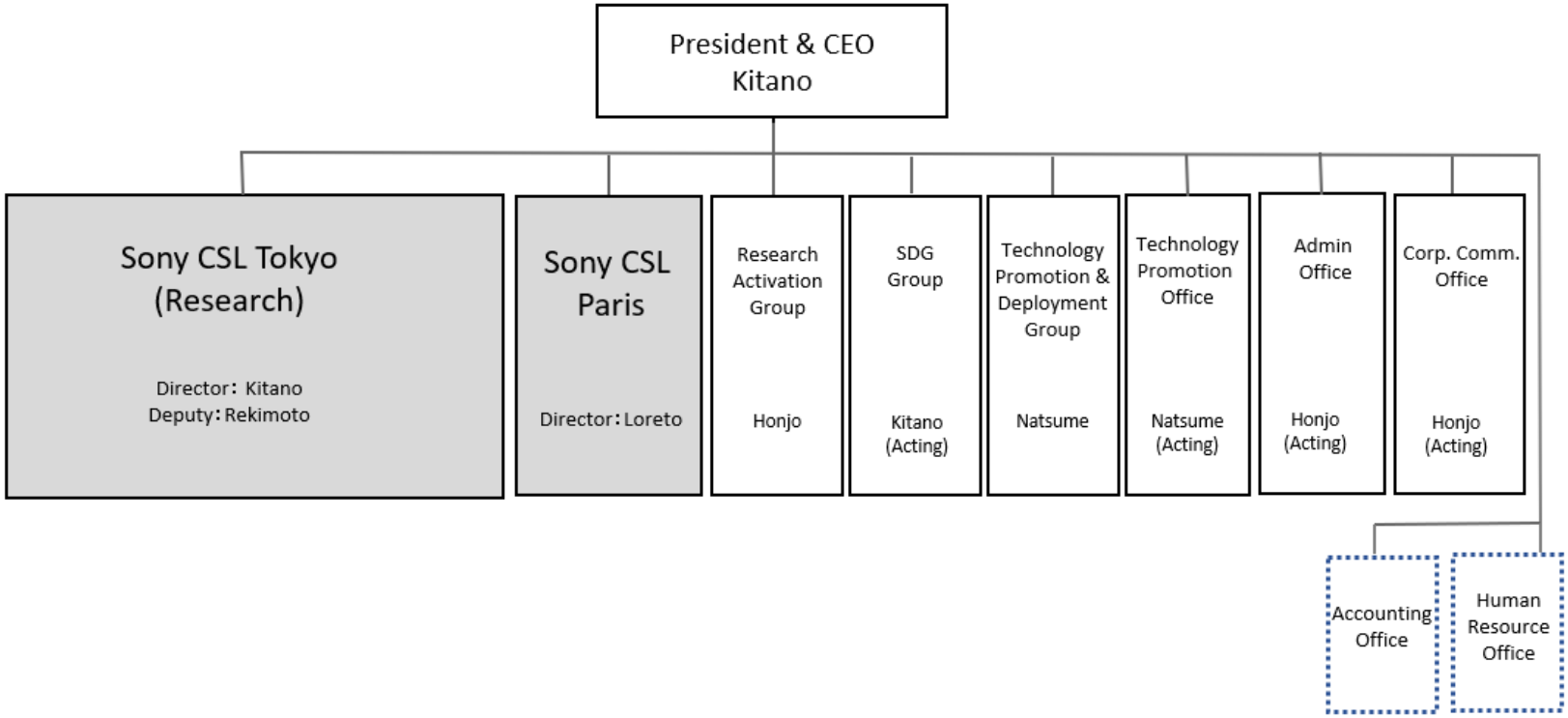
Kitano says:

> *My major preoccupation has always been how to find far-reaching solutions to major problems and what frameworks must be created to accelerate progress toward those solutions. I studied particle physics in college, later moving into computer science, in particular, artificial intelligence research, and then engaging in research into massively parallel artificial intelligence, spoken-language translation systems, and machine learning. But I felt that in order to accelerate progress in AI and robotics, it would be necessary to establish some kind of big goal and global project. This became RoboCup, a project with the target of developing a team of fully autonomous humanoid robots able to win the FIFA World Cup by 2050. At the same time, I started to realize that intelligence is a byproduct of evolution, so I would have to study life itself. And so I switched to biosciences research, where I soon recognized the need for a conceptual shift from reductionist research into genes and proteins to a systems thinking approach. In order to accelerate that shift, I advocated "systems biology," an emerging area of life science back in the mid '90s, which is a well-accepted discipline in life science today. At the same time, it became clear that human intelligence may fall short of handling the vastness and complexity of living systems and data generated to uncrack the system. An artificial intelligence system must be developed that can assist humans in scientific research and accelerate the speed of scientific discovery. This may imply the modality of scientific research itself may be transformed into Human-AI symbiotic activities. To push this idea, I have recently formulated the "Nobel Turing Challenge," a global grand challenge to develop a highly autonomous "AI Scientist" to*

make major scientific discoveries, some of which may be worthy of the Nobel Prize. I believe this research involved in this challenge to be of the highest scientific and technological importance, with the power to reshape our civilization.

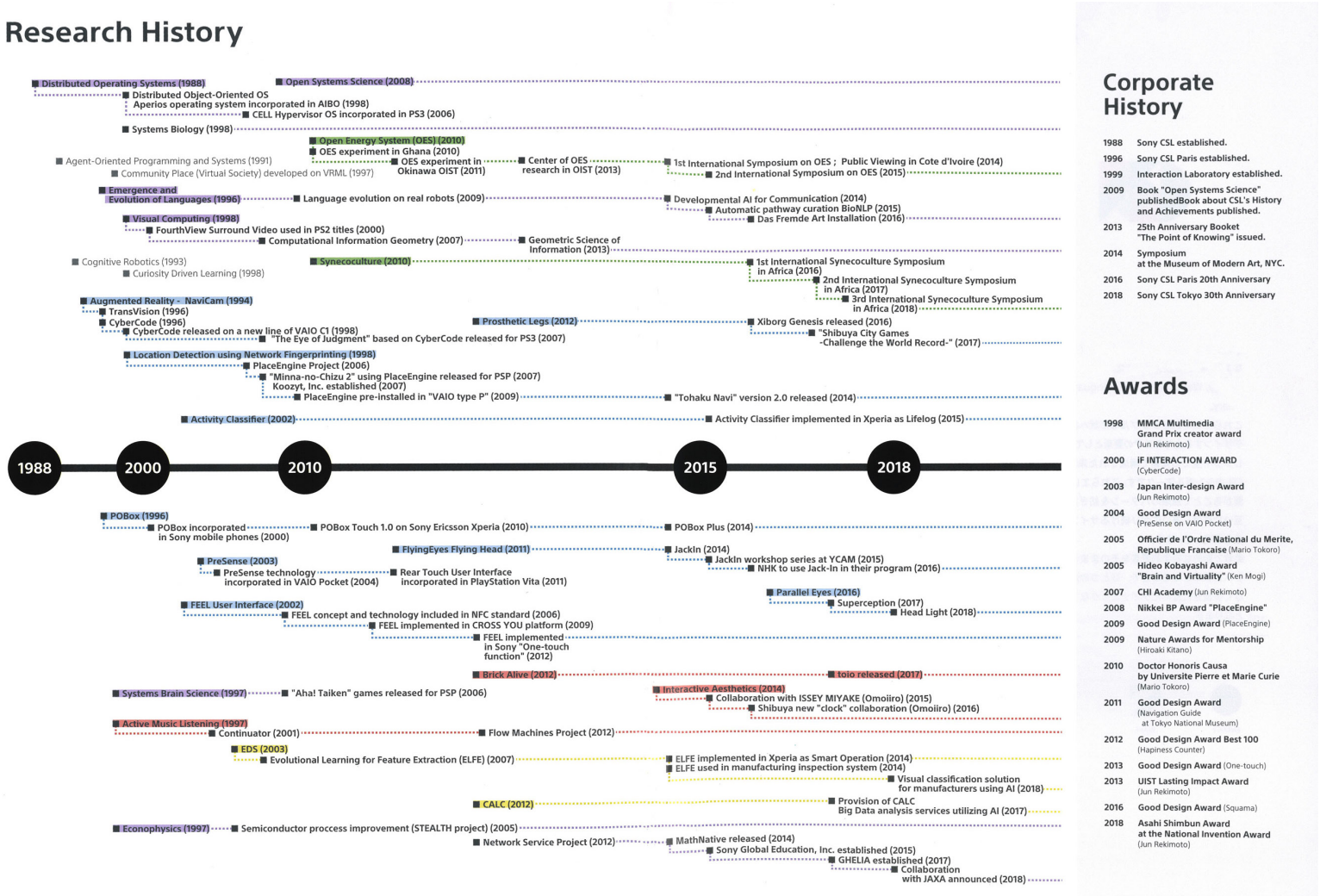
Source: Sony CSL, "Hiroaki Kitano," accessed March 31, 2020, https://www.sonycsl.co.jp/member/tokyo/176; Kitano, interview with case authors.

Exhibit 5: Sony CSL Organization Chart, 2019



Source: Sony CSL.

Exhibit 6: Research Projects of Sony CSL



Source: Sony Computer Science Laboratories marketing materials.

Exhibit 7: Details of Selected Sony CSL Research Projects

Econophysics

The research of Hideki Takayasu, a PhD in physics who began working at Sony CSL in 1997, tackled the analysis of economic data using methodologies originally developed for studying physics, in a new scientific field called econophysics.[32] Econophysics is based on the idea that the properties and dynamics of socioeconomics systems can be modeled using principles, formulae, and empirical observations used in the field of physics.[33] Specifically, Takayasu applied mathematical models from physics and statistics to economic phenomena. Among other things, his work measured stochastic influences (i.e., having a random probability distribution) on the Japan and US stock and foreign exchange markets; used the principles of Brownian motion to analyze yen-dollar exchange rates; and demonstrated that market prices exhibit similar statistical properties to an electrical circuit that hovers between damping and amplifying phases.[34]

Takayasu's econophysics work in statistical analysis resulted in the development of a database system that contributed to faster development and improved yields of semiconductors. The bottom-line benefit to Sony of Takayasu's research equaled tens of millions of dollars per year.[35]

Musical Dynaformics

Shinichi Furuya, an accomplished pianist with a PhD in Medical Sciences, had been researching the connection between body movement and creative output at Sony CSL since 2017. With his team of postdoctoral fellows and other students, he conducted pioneering research in the field of musical dynaformics, which combined neuroscience, robotics, biomedical engineering, and music theory.

Becoming a virtuoso requires development of both technical skills, which involve biological processes like speed and accuracy of finger movements, and interpretation, an unconscious instinct for transmitting a mood with sound and tone. An expert pianist has such a command over his finger movements that he can use his body's movements to express a certain quality or element of the music he is performing. It takes thousands of hours of practice to develop these skills, and yet they are difficult to teach. Furuya's goal was to help musicians achieve new peaks of creative expression by optimizing their training and practice methods.[36]

In one of Furuya's projects, an expert pianist wore a "data glove" equipped with microsensors that detected and transmitted data on the movements of his fingers, while an amateur pianist wore an exoskeleton receiving input from the glove. This exoskeleton reproduced, in real time, the movement of the expert's fingers, allowing the amateur to intuitively learn how to play better by mimicking the exact motions of an expert without conscious effort. Furuya hoped these robotics-enhanced tools could enhance the way music is learned and practiced, making it more efficient to master the technical aspects and leaving more time to cultivate personal expression and style, thus elevating the art form.

For more, see "Beyond the Limits of Musical Expression Through Technology," posted November 10, 2019, https://youtu.be/GB8wtjmOoBY.

Small Optical Link

A team of researchers, software developers, and engineers, headed by Taiji Ito of the TPO, had been working since 2016 with JAXA, the Japan Aerospace Exploration Agency, to develop a long-distance real-time communication system using laser technology. The idea was born of Sony's existing optical disk technology, first developed in the 1970s, which had been used to transmit data in dozens of Sony's consumer devices, including CD players, DVD players, and PlayStations. Merging this old technology with new technology, the team developed a system that could transmit data over distances of up to 4,500 kilometers (2,800 miles), further than previously possible. This laser communication system, when installed on a satellite, would be able to transmit data to other satellites as well as with ground stations on Earth.

The partnership with JAXA enabled the team to use the International Space Station for in-orbit testing, which accelerated the rate of research and development for the project. The Small Optical Link for the International Space Station (SOLISS) laser communication system was launched for in-orbit testing in September 2019. After the conclusion of and the follow-up to the testing, Sony CSL hoped to commercialize this technology.[37] Real-time data transmittal from space was not yet possible, but eventually this technology could be used to set up a communication network between the International Space Station, Earth, the moon, and Mars. Yet Sony CSL's focus was not only on space; it saw vast social applications in enabling real-time broadband communication around the globe, such as in regions that did not have Internet access.[38]

For more, see "Why Is Sony's Technology Heading to Space?" posted November 27, 2019, https://youtu.be/4DMZKSCqqbw.

Synecoculture

In 2010, researcher Dr. Masatoshi Funabashi launched a new field of science, called Synecoculture™, while working at Sony CSL. Synecoculture, in short, was a new approach to farming that brought together insights from a variety of disciplines—ecology, information science, complex systems science, physiology, environmental science—with the goal of optimizing food production, quality, and ability to adapt to climate change.

Conventional agriculture often saw small-scale farmers trade biodiversity, which is essential for food production, for short-term productivity, damaging the soil's fertility and having far-reaching consequences for not just food production but the Earth's entire ecosystem. Further, conventional agriculture historically focused on growing only a small fraction of the 30,000 known species of crops that have useful applications for society. Funabashi's work used big data and AI to interpret the complex systems involved in agriculture and help make decisions on which plant combinations worked best together.[39]

The methods Funabashi and his team developed allowed for a thriving ecosystem without tilling, fertilization, or pesticides.[40] The potential for this field of research was enormous, as reports warned that current agricultural practices had caused irreversible damage to our ecosystem.[41] The increase in rich, biodiverse farms that synecocultural practices could provide might mean a radical transformation in agriculture that could have a global impact for future generations.

For more, see "Renewing Agriculture to Rebuild a Broken Environment [at] Sony Computer Science Laboratories, Inc.," posted September 27, 2017, at https://youtu.be/cQEr0vN7guM.

Exhibit 8: Comparing Corporate-Funded Fundamental Research Labs

	Sony CSL	IBM Watson Research Center	X	MIT Media Lab
Structure/ Governance	100% subsidiary of Sony; all researchers report directly to president, who reports directly to Sony CEO	Hub-and-spoke model; WRC was headquarters overseeing 12 labs on 6 continents	Started as unit within Google; spun off into separate company with its own head in 2015; brought back under Google umbrella in late 2019	25 research groups composed of students with a faculty member or senior researcher leading them; each group's leader reported to the lab's director, who reported to MIT president and MIT governing board
Routes to Impact	TPO and TPDG offices within Sony CSL, staffed by non-scientists, facilitated tech transfer, licensing, and external partnerships; Research Activation Group (RAG) supported academic and social impact projects of high importance	Classified emerging business opportunities, which triggered different management and evaluation processes; goal was to transition H3 projects into H2, with more predictable revenue streams	X considered more than 100 ideas each year, but only a fraction became projects with full-time staff dedicated to them; several X projects had "graduated" into separate business entities under the Alphabet umbrella, but none had made money yet	Member companies collectively owned the IP on Media Lab research in perpetuity, allowing them to commercialize the technology if desired
Funding	100% funded by Sony	100% funded by IBM; budget $5B	Not publicly reported; possibly $1B	$70M, 80% supplied by corporate sponsors and remainder from MIT
Size and Location	30 principal researchers in Tokyo and Paris with small supporting administrative staff	3,000 employees worldwide	Not publicly reported; located in Mountain View, CA	30 faculty and senior researchers; over 175 research associates; and about 100 administrative staff in Cambridge, MA
Scope	Unlimited; project theme proposed by researcher and approved by director	Must have reasonable, attainable goals with defined business purpose and strategic intent	Must have potential for global impact; must be achievable in 5–10 years; must be "audacious"[42]	Ideas must live in the "white space" between fields of study; driven by discovery, not profit or business use

Sources:

Sony CSL: Kitano, interview with case authors.

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X: Derek Thompson, "Google X and the Science of Radical Creativity," *The Atlantic*, November 2017, https://www.theatlantic.com/magazine/archive/2017/11/x-google-moonshot-factory/540648; Astro Teller, "A Peek Inside the Moonshot Factory Operating Manual" (blog post), July 23, 2016, https://blog.x.company/a-peek-inside-the-moonshot-factory-operating-manual-f5c33c9ab4d7; Mark Bergen, "The Google X Moonshot Factory Is Struggling to Get Products Out the Door," *Vox*, August 29, 2016, https://www.vox.com/2016/8/29/12663630/google-x-alphabet-moonshot.

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Exhibit 9: Different Possible Approaches to Performance Reviews

1. Deferred Evaluation

	President of Sony CSL	Researcher
Project Start	Together, the Sony CSL president and the researcher determine the project scope and evaluation criteria. They establish a midterm goal for the project that should be achieved within 3 years from now.	
1 year after project start	Reports to Sony's CEO and CFO on the project's progress. No formal evaluation done.	Updates the Sony CSL president on the project's progress. President offers advice and redirection, if needed, but no formal evaluation done.
3 years after project start	Reports to Sony's CEO and CFO on the project's progress. No formal evaluation done.	Submits to midterm review that analyzes whether researcher achieves the 3-year goal of the project. President will decide whether Sony CSL will continue funding.
10 years after project start	Is formally evaluated on the project's success by a committee comprising Sony's CEO, CFO, and board. Criteria for evaluation might include (1) direct profit to Sony, (2) impact on the world, and (3) level of media attention.	Is formally evaluated on the project's success by president of Sony CSL and an independent project review committee. Criteria for evaluation will focus more on the impact the technology has had on the world, and less on the direct benefit to Sony.

2. Annual Evaluation, with Goals Defined by Researcher

	President of Sony CSL	Researcher
Project Start	The researcher determines the project scope, evaluation criteria, and goals for project after 1, 3, and 10 years. The Sony CSL president reviews and approves (or tweaks) these goals.	
1 year after project start	Is formally evaluated by Sony's CEO and CFO on the project's success in meeting its 1-year goal, balanced with its strength when measured in the context of the entire Sony CSL project portfolio.	Is formally evaluated by Sony CSL president on the project's success in meeting its 1-year goal. President will decide whether Sony CSL will continue funding.
3 years after project start	Is formally evaluated by Sony's CEO and CFO on the project's success in meeting its 3-year goal, balanced with its strength when measured in the context of the entire Sony CSL project portfolio.	Is formally evaluated by Sony CSL president on the project's success in meeting its 3-year goal. President will decide whether Sony CSL will continue funding.
10 years after project start	Is formally evaluated by Sony's CEO and CFO on the project's success in meeting its 10-year goal, balanced with its strength when measured in the context of the entire Sony CSL project portfolio.	Is formally evaluated by Sony CSL president on the project's success in meeting its 10-year goal.

3. Annual Evaluation, with Goals Defined by Evaluation Committee

	President of Sony CSL	Researcher
Project Start	After interviews with Sony CSL president and researcher about the project and its goals, an independent evaluation committee establishes evaluation criteria and goals for project after 1, 3, and 10 years.	
1 year after project start	Is formally evaluated by the committee on the project's success in meeting its 1-year goal, balanced with its strength when measured in the context of the entire Sony CSL project portfolio. Committee sends its report and recommendations to Sony's CEO and CFO.	Is formally evaluated by Sony CSL president and evaluation committee on the project's success in meeting its 1-year goal. Evaluation committee will decide whether Sony CSL will continue funding.
3 years after project start	Is formally evaluated by the committee on the project's success in meeting its 3-year goal, balanced with its strength when measured in the context of the entire Sony CSL project portfolio. Committee sends its report and recommendations to Sony's CEO and CFO.	Is formally evaluated by Sony CSL president and evaluation committee on the project's success in meeting its 3-year goal. Evaluation committee will decide whether Sony CSL will continue funding.
10 years after project start	Is formally evaluated by the committee on the project's success in meeting its 10-year goal, balanced with its strength when measured in the context of the entire Sony CSL project portfolio. Committee sends its report and recommendations to Sony's CEO and CFO.	Is formally evaluated by Sony CSL president and evaluation committee on the project's success in meeting its 10-year goal.

Source: Created by the authors.

Endnotes

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